



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	STATISTICAL MECHANICS				
Type of Course	Core in Major				
Semester	VII				
Academic Level	400 – 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	A solid foundation in classical mechanics, quantum mechanics, and thermodynamics. Additionally, proficiency in calculus, differential equations, and linear algebra is essential for understanding the mathematical formalism used in statistical mechanics. A familiarity with probability theory and basic concepts of probability distributions can also be beneficial, as statistical mechanics involves the statistical analysis of large ensembles of particles to understand their collective behavior and properties.				
Course Summary	The course on statistical mechanics explores the principles governing the collective behaviour of large systems of particles, utilizing probabilistic methods to understand thermodynamic properties and the microscopic origins of macroscopic phenomena.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the concept of multiplicity	Understand	Conceptual Knowledge	Written exams, quizzes
CO2	Apply the second law of thermodynamics	Apply	Procedural Knowledge	Problem sets, lab experiments
CO3	Analyze changes in entropy in various systems	Analyze	Conceptual & Procedural Knowledge	Case studies, simulations
CO4	Utilize Boltzmann statistics in statistical mechanics	Apply	Conceptual & Procedural Knowledge	Problem-solving exercises, projects
CO5	Employ quantum statistics in understanding systems	Apply	Conceptual & Procedural Knowledge	Research papers, presentations
CO6	Evaluate thermodynamic variables in complex systems	Evaluate	Conceptual & Procedural Knowledge	Research projects, oral exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	MULTIPLICITY, ENTROPY AND THE SECOND LAW		11	16
	1	Two-State Systems, the Two-State Paramagnet	2	

	2	The Einstein Model of a Solid	1	
	3	Interacting Systems, Large Systems, Stirling's Approximation	2	
	4	Multiplicity Function of a Large Einstein Solid and Its Sharpness	2	
	5	Multiplicity Function of a Monatomic Ideal Gas and Interacting Ideal Gases	2	
	6	Multiplicity and Entropy, Entropy of an Ideal Gas, Entropy of Mixing, Reversible and Irreversible Processes	2	
	Sections 2.1 – 2.6 of chapter 2 of Book 1			
II	INTERACTIONS, CHANGE IN ENTROPY AND THERMODYNAMIC VARIABLES		10	16
	7	Thermal Equilibrium and Temperature	2	
	8	Change in Entropy and Heat Capacity, Measuring Entropy, Macroscopic View of Entropy	2	
	9	Paramagnetism: Analytic Solution only (The numerical solution of this problem is included as one of the experiments)	2	
	10	Mechanical Equilibrium and Pressure, Entropy and Heat Revisited	2	
	11	Diffusive Equilibrium and Chemical Potential	2	
	Sections 3.1 – 3.6 of chapter 3 of Book 1			
III	BOLTZMANN STATISTICS		10	16
	12	The Boltzmann Factor, the Partition Function and Average Values	2	
	13	Paramagnetism Revisited Using the Partition Function	1	
	14	Equipartition Theorem	1	
	15	The Maxwell Speed Distribution	2	
	16	Partition Functions and Free Energy	1	

	17	Partition Functions for Composite Systems	1	
	18	Ideal Gas Revisited Using the Partition Function, Predictions	2	
	Sections 6.1 – 6.7 of chapter 6 of Book 1; Problems in chapters 5, 19 and 20 of Book 2			
IV	QUANTUM STATISTICS		14	22
	19	The Gibbs Factor	1	
	20	Bosons and Fermions, The Distribution Functions	2	
	21	Degenerate Fermi Gases: Zero Temperature, Small Nonzero Temperatures, The Density of States, Sommerfeld Expansion	3	
	22	Blackbody Radiation: The Ultraviolet Catastrophe, The Planck Distribution, Photons, Summing over Modes, The Planck Spectrum, Total Energy, Entropy of a Photon Gas, Cosmic Background Radiation, Photons Escaping through a Hole, Radiation from Other Objects, The Sun and the Earth	4	
	23	Debye Theory of Solids	2	
	24	Bose-Einstein Condensation, Real-World Examples, Why Does It Happen?	2	
	Sections 7.1 – 7.6 of chapter 7 of Book 1; Problems in chapters 21, 23, 24, 29 and 30 of Book 2			
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.			
	1	Variation of surface tension with temperature - Jaeger's method. To determine the surface tension of water at different temperatures by Jaeger's method of observing the air bubble diameter at the instant of bursting inside water.		

2	Stefan's constant - To determine Stefan's constant.		
3	Thermal conductivity of liquid and air by Lee's disc method.		
4	Viscosity of a liquid - Oscillating disc method. To determine the viscosity of the given liquid by measurements on the time period of oscillation of the disc in air and in the liquid.		
5	Measurement of the thermal and electrical conductivity of Cu to determine the Lorents number.		
6	Curie Weiss law - To determine the Curie temperature.		
7	Measurement of the thermal relaxation time constant of a serial lightbulb.		
8	Simulate the time dependent positions of collection of particles, having initial random velocity distribution, confined to a one dimensional box (See section 2.1 in Book 3).		
9	Simulate the Statistical behaviour of two Einstein solids, (Solid A contains 200 oscillators and solid B contains 300 oscillators) sharing a total 100 units of energy, that can exchange energy. Find the equilibrium of the systems (See section 3.1 in Book 1).		
10	Simulate the entropy, temperature and heat capacity of an Einstein solid containing 50 oscillators (initially) and from 0 to 100 units of energy (See problem 3.24 in Book 1).		
11	Simulate the statistical behaviour of two-state paramagnet (spin half system) (see section 3.2 in Book 1).		
12	Simulate the statistical nature of the Boltzmann distribution by distributing quanta of energy in a lattice of size 20*20 and plotting the histogram. Track the variation of the number of microstates (see example 4.2 in Book 2).		
13	Simulate the statistics of occupation number (distribution function) of an ideal, quantum mechanical, non interacting i) Maxwell-Boltzmann ii) Bose-Einstein and iii) Fermi-Dirac systems.		
14	Simulate the temperature dependent average energy per particle of an n-level system in thermal equilibrium at various temperatures for n = 2, 3, 4, 10 and 30 or higher. Use a normalized axis for comparison and draw the conclusions (see example 20.3 in Book 2).		
15	Simulate the temperature dependent heat capacity per particle of an n-level system in thermal equilibrium at various temperatures for n = 2, 3, 4, 10 and 30 or higher values. Use a normalized axis for comparison and draw the conclusions (see example 20.3 in Book 2).		

16	Simulate the temperature dependent average energy, entropy and heat capacity of the harmonic oscillator in thermal equilibrium at various temperatures. Use a normalized axis for comparison and draw the conclusions (see example 20.3 in Book 2).		
17	Simulate the temperature dependence of fermi energy. Also simulate the dependence of number density and temperature on ground state pressure of an ideal fermi system.		
18	Simulate the black body radiation spectra for three different temperatures and demonstrate the Wein's displacement law and Stefan's law (see section 23.6 in Book 2).		
19	Simulate the temperature dependent heat capacity of an ideal Bose in the temperature range from 0 K to a high temperature four times the Bose-Einstein condensation temperature . Use a normalized axis for comparison and bring out the feature of Bose-Einstein condensation.		
20	Simulate the temperature dependent pressure of an ideal Bose in the temperature range from 0 K to a high temperature four times the Bose-Einstein condensation temperature. Use a normalized axis for comparison and bring out the feature of Bose-Einstein condensation.		
21	Simulate the behaviour of a quantum mechanical paramagnetic system as a function of B/T for systems with spins half to infinity (classical limit).		

Books and References:

1. Introduction to Thermal Physics (Oxford Edn., 2021) by Daniel V Schroeder (Book 1)
2. Concepts in Thermal Physics (Oxford Edn., 2006) by Stephen J Blundell and Katherine M. Blundell (Book 2)
3. Thermal Physics Tutorials with Python Simulations (CRC Press, 2023) by Minjoon Kouh and Taejoon Kouh
4. Fundamentals of Statistical and Thermal Physics by Frederick Reif
5. Statistical Mechanics by R.K. Pathria and Paul D. Beale
6. Equilibrium Statistical Physics by Michael Plischke and Birger Bergersen
7. Python for Education by Dr. B P Ajithkumar, IUAC, New Delhi; e-book freely downloadable from <https://scischool.in/python/index.html>

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	3	1	1	0	3	1	2	1	1	1	3	1	0
CO 2	3	2	2	0	3	1	2	1	1	1	3	1	0
CO 3	3	2	3	0	3	1	2	1	1	1	3	1	0
CO 4	3	1	3	1	3	1	3	1	1	2	3	1	1
CO 5	3	1	3	1	3	1	3	1	1	2	3	1	1
CO 6	3	0	3	1	2	1	3	0	1	2	3	1	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	